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RESEARCH ARTICLE

Proteome-Scale Mining of Metal-Associated Proteins of Monkeypox Virus

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ABSTRACT

Metal ions are crucial for viral processes like replication, structural integrity and immune modulation, despite that the metalloproteome of Monkeypox virus (MPXV) remains largely unexplored. Monkeypox virus is a re-emerging zoonotic Orthopoxvirus with a 197 kb genome encoding over 183 proteins. Here, in this report we aimed to identify and explore the metal-associated proteome of the MPXV. Derived from employing a structure driven pipeline, followed by the functional annotation, the subcellular localization, evolutionary aspect, and structural validation with the established experimental evidences. Yielded a set of approximately 21 % high-confidence putative metal-associated proteins with a potential as metal-binding proteins. Functional annotation, Gene Ontology (GO) enrichment, and KEGG Orthology (KO) term assignment also revealed significant enrichment of pathways primarily related to functions such as, viral replication, genome maintenance, transcription, virion assembly, and host immune modulation. These metal-associated proteins are hypothesized to perform critical biological roles throughout the viral life cycle and pathogenesis. This includes nucleotide metabolism, transcriptional regulation, redox balance and immune evasion, and virion morphogenesis. These findings were further strengthened by subcellular localization analysis, which predicts the presence of metalloproteins within MPXV and its viral factories. Further, suggesting the spatial distribution of various metals and its utilization in the host organism. Facilitating the activities from viral attachment, entry, replication, transcription, viral assembly, and release as either mature virion (MV) or intracellular mature virion (IMV). Comparison and mapping these identified metal-associated MPXV proteins to Vaccinia virus followed by the virus-host network analysis highlighted the proposed role of metal-associated proteins within conserved Orthopoxvirus interaction pathways.

1 | Introduction

The mpox virus (MPXV), is an enveloped, double stranded DNA virus, which belong to the Orthopoxvirus genus, and family Poxviridae. MPXV has a large genome (~197 Kb) that encodes an extensive collection of 183 proteins, many dedicated to evading host immune response and hijacking host cellular machinery for replication. It has emerged from a neglected zoonotic disease namely mpox which earlier remain confined to Central and Western Africa to a significant global public health threat since the 2022 outbreak. The 2022 outbreak of mpox continues to this day and was even declared as a Public Health Emergency of International Concern (PHEIC), affecting several other countries, and undersestimated its capacity for sustained human to -human transmission and its establishment and spread in non-endemic regions [1, 2]. While there are therapeutic options like antiviral medications (Tecovirimat, Cidofovir, or Brincidofovir) available for its management and vaccination for its prevention, these measures are not particularly meant for mpox, revealing a critical and possible gap in our understanding of this virus and its fundamental molecular virology [3].

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Significance of the study

Metal ions are central to many viral processes, yet their degree of involvement and dependency in MPXV biology and pathogenesis has not been previously explored. One of the primary reasons could be due to the fact that because the direct experimental characterization of any viral metalloproteome may involve significant technical challenges. As extraction, purification and crystallization of these metal-associated proteins requires stringent conditions to maintain the steady metal coordination structure. Where during the sample processing these metal ions often dissociate or get exchanged with native metals, resulting in the formation of apo structures. Whereas these apo structures may not represent the true native structure of the metal bound proteins. Therefore, the entire process of experimental characterization of metal associated proteins across the viral proteome can be technically challenging and resource intensive. Our study addresses this limitation and by adopting an integrated computational and established experimental evidence based structural validation approach. This study delivers the proteome-wide, structure-validated assessment of candidate MPXV metal-associated proteins, and their possible role across the various stages of the viral life cycle and pathogenesis. Our findings on these viral metal-associated proteins can facilitate to understand the currently experimentally unexplored proteins for future therapeutic interventions, and development of antivirals for mpox.

Metal ions are an integral part of many proteins, supporting and enhancing their structure and function across all the domains of life, including viruses. They act as cofactors which are crucial for various cellular activities such as enzymatic catalysis, structural stabilization, nucleic acid binding, and redox regulation [4]. Viral genomes, despite their compactness, often encode metal-associated proteins that depend on transition metals such as zinc, magnesium, manganese, iron, copper, and cobalt to perform and maintain their stable structure and often support their fundamental activities like replication, transcription, and virion assembly [5]. The dependence of viruses on the metal ions gained from the host organism, highlights their role as critical molecular modulator, determining its virulence [6].

Therefore, a systematic approach in understanding these complex metalloproteome of these zoonotic viruses is of an utmost importance to understand its overall biology. However, studying these metalloproteins serves major challenges during extraction purification and characterization *in-vitro*. The major reason being, as these metal ions may often gets dislodged due to changes in ionic strength or due to the presence of chelating agents and thus requires careful execution making it a resource-intensive process [7, 8]. With the current advancement in structural biology and computational annotation, systematic study of viral metalloproteomes has revealed that a substantial fraction of viral proteins contains conserved metal-binding motifs [6, 9]. These motifs frequently coincide with important catalytic or nucleic acid-binding domains, suggesting that metal binding is a universal strategy employed by these viruses. Which may be used by them to enhance their protein stability and performance under limited coding capacity during infection [5, 10].

Recent studies utilizing bioinformatics tools and computational approaches have revealed the importance of these metal-associated proteins in RNA and DNA viruses and their suggestive role in viral-host interactions [5, 6]. In addition, these metal-associated proteins have been identified as druggable targets by metal chelators and ionophores. Certain chemical compounds like Disulfiram and Ebselen have been experimentally identified to be potential "zinc-ejectors" as they can destabilize zinc-thiolate clusters in viral proteins. Ultimately leading to enzyme inactivation and inhibition of replication in SARS-CoV-2 and other viruses [11]. These findings reinforce that metal-associated proteins are not only central to viral pathogenicity but may also represent potential antiviral targets.

In this study, we used a computational-based approach to perform a comprehensive proteome-wide identification and characterization of potential metal-associated proteins in MPXV. Integrating AlphaFold3 (AF3) structure prediction based on ipTM & pTM confidence metrics with functional annotation, followed by subcellular localization predictions, and ortholog mapping to the Vaccinia proteome. It was observed that numerous MPXV proteins have binding affinity to metal ion, including zinc, magnesium, manganese, iron, copper, and cobalt. Many of which map to show conserved enzymatic and regulatory domains which are essential for viral replication inside the host organism. We validated these computational predictions by performing BLASTp on 39 candidate metal associated MPXV proteins against experimentally solved PDB (Protein Data Bank) homologs, confirming sequence conservation of metal coordinating residues among orthologous proteins. The top PDB homolog was then structurally aligned with the AF3 predicted structure using TM-align structural superimposition, assessing alignment quality via TM-score and RMSD. This two-step validation not only strengthens the prediction confidence by confirming that metal-interacting residues are evolutionary and structurally conserved, providing a rational hypothesis for experimental and biochemical validation.

2 | Results

2.1 | Identification of Candidate Metal-Associated Proteins in MPXV

In this study, the complete reference proteome of Monkeypox virus (UP000516359) was retrieved from the UniProt proteome database, comprising 183 proteins. On downstream homology-based screening using standalone BLASTp (Standard parameters; *e*-value $< 1 \times 10^{-5}$, and percentage identity $> 50\%$) against curated metal-binding protein datasets of transient metals, including zinc (33,399 proteins), magnesium (25,848), manganese (6,647), iron (29,421), copper (33,673), and cobalt (1,215) were obtained from the UniProtKB.

The metal-wise BLASTp results were manually checked to remove redundancy and duplicate entries, yielding 35 zinc, 30 magnesium, 16 manganese, 15 iron, 9 copper, and 5 cobalt-associated hits. To further validate and refine the putative

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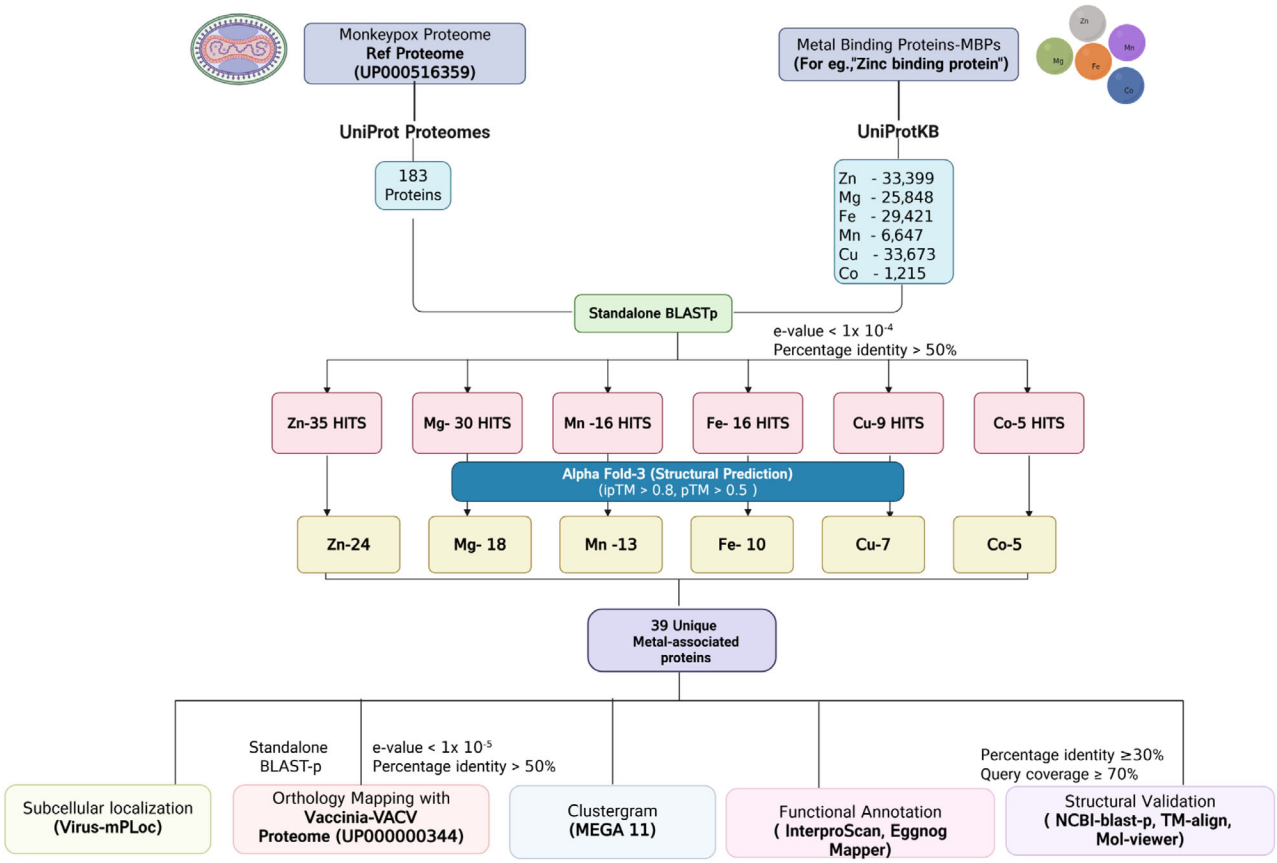


FIGURE 1 | Overview of the computational pipeline for identification and characterization of candidate metal-associated MPXV proteins.

metal-associated proteins, their three-dimensional structures were predicted using AF3. Proteins with high-confidence scores (ipTM > 0.8 and pTM > 0.5) were shortlisted (Table S1). These candidates were further analyzed and cross-compared to remove overlapping entries, as shown in Figure 1. A total of 39 high-confidence metal-associated proteins, collectively representing the putative MPXV metalloproteome, were selected for downstream functional annotation and pathway analysis. Figure 2 illustrates the binding affinities of all the identified high-confidence putative metal-associated proteins highlighting their potential to bind to more than one metal. Notably, among these identified 39 metal-associated MPXV proteins, 19 proteins showcased affinity for more than one metal ion. For an instance, A0A7H0DN21 (ribonucleoside-diphosphate reductase small subunit) exhibited binding potential for four different metal ions such as, zinc, manganese, iron and cobalt ions. Whereas certain protein molecule A0A7H0DNE6 (DNA ligase) showed affinity only for single metal ion cobalt (Supporting information S2).

2.2 | Distribution of MPXV Proteins Among Zinc, Magnesium, Manganese, Iron, Copper, and Cobalt

The predicted MPXV metalloproteome was dominated by predominantly proteins associated with zinc (31%), followed by magnesium (23%) and manganese (17%). In addition, iron-associated proteins accounted for 13%, while copper 9% and cobalt (7%) constituted a smaller fraction of the viral metalloproteome Figure 1.

This distribution shows the preponderance of magnesium and zinc as their principal metal cofactors. These findings remain consistent with their essential roles in viral replication, transcriptional regulation, and protein stabilization. In addition, the predicted association with other metal ions such as manganese, iron, copper, and cobalt, MPXV proteins further exhibit their metal-dependent role in activities like, redox regulation, enzymatic catalysis, and host-virus interaction.

2.3 | Functional Annotation, GO Enrichment Analysis of Predicted MBPs

Functional annotation of the 39 shortlisted metal-associated proteins of MPXV was performed using InterProScan v5.76-107.0 to identify the conserved domains and their respective protein families.

Domain analysis revealed a diverse set of conserved protein families, especially the metal-dependent enzymatic activity, redox regulation, and nucleotide-binding functions. Further classification based on protein families and domains identifies that these proteins are involved in enzymatic catalysis, protein-protein interactions and metal coordination (Supporting information S2, for detailed annotation and functional classification). It has also been observed that most of these candidate metal associated proteins include conserved cysteine-rich regions and catalytic fold in their structure.

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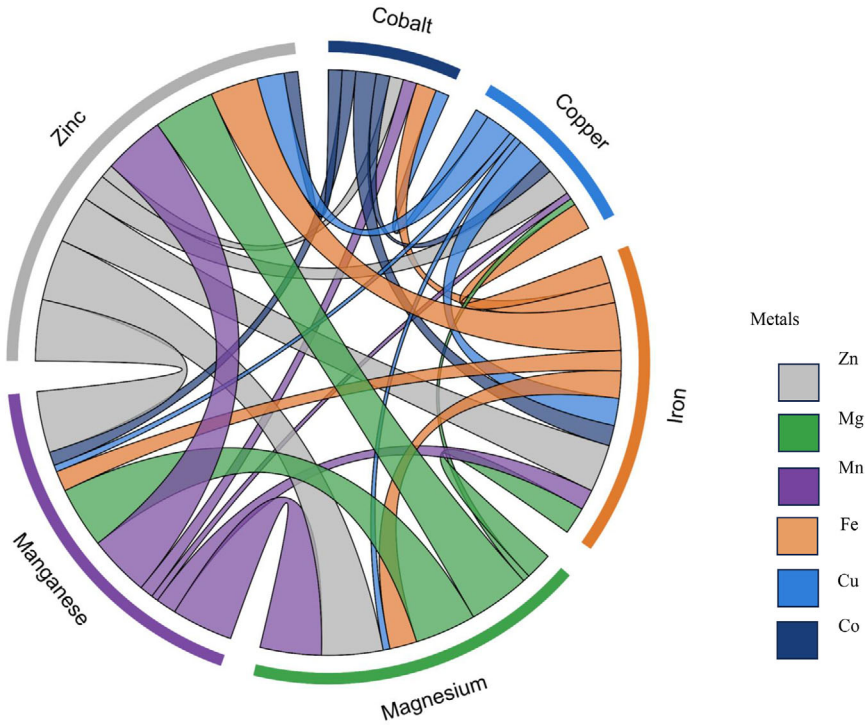


FIGURE 2 | Predicted metal-binding affinities of 39-candidate MPXV proteins to bind Zn, Mg, Mn, Fe, Cu & Co using AF3 predictions thresholds (ipTM > 0.8 and pTM > 0.5). This figure summarizes the metal ion affinities of 39 candidate MPXV metal-associated proteins predicted through sequence homology analysis. Detailed AF3 (ipTM & pTM)metrics are provided in Table S1.

GO enrichment analysis of the 39 MPXV metal-associated proteins also revealed consistent enrichment across biological processes (BP), cellular components (CC), and molecular functions (MF) categories. In the biological processes category, the most enriched terms were DNA recombination (GO:0006310), deoxyribonucleotide biosynthetic process (GO:0009263), and DNA repair (GO:0006281), indicating a strong association of these metal-associated proteins with viral genome replication and maintenance (Figure 3A). Cellular component analysis showed predominant enrichment for host cell (GO:0030430), followed by host cell Golgi apparatus (GO:0044177) and virion component (GO:0044423), consistent with cytoplasmic replication and virion-associated functions of MPXV proteins (Figure 3B). In the molecular function category, the most enriched terms included metal ion binding (GO:0046872), Zinc ion binding (GO:0008270), and nuclease activity (GO: 0004518), further strengthening the metal-dependent and enzymatic nature of the proposed MPXV metal-associated proteins (Figure 3C).

To further contextualize the GO enrichment results, KEGG Ortholog (KO) assignment was performed for the 39 MPXV metal-associated proteins. 30 MPXV metal-associated proteins were assigned successfully to KO terms, while 9 remain unassigned. The assigned KO terms supported the GO enrichment results, with MPXV proteins involved in deoxyribonucleotide biosynthesis and DNA recombination mapping to KOs associated with DNA precursor production and genome repair (K10807, K10808, K10776, and K25268). Proteins enriched for virion components and host cytoplasm correspond to KOs linked to viral structural assembly and intracellular metabolic maintenance (K25273, K01054, and K16860). At the molecular level, enrichment of zinc and metal ion binding mapped to KOs associated with redox regulation and catalytic enzymatic activity (K16860, K01672, K00857, and K10807). Details of KO assignment are given in Supporting information S4.

Based on the above functional annotation, the proposed metal-associated proteins are primarily involved in key viral functions that contribute to monkeypox virus (MPXV) pathogenesis, including DNA replication, repair, transcriptional machinery, post-translational modification, immune evasion, and host range determination. Together, these processes coordinate efficient viral replication, modulation of host responses, and successful infection (Figure 4A). This functional framework enabled the proposed categorization of MPXV metal-associated proteins into five major functional groups: (i) viral replication enzymes involved in DNA/RNA synthesis and repair (ii) viral host interaction and immune modulation, (iii) viral protein processing and post translational modification, (iv) viral assembly and structural proteins and (v) accessory proteins facilitating viral replication (Figure 4B).

2.4 | Subcellular Localization and Cluster Analysis

Furthermore, subcellular localization of viral proteins is critical for understanding their functional roles during infection and their contributions to viral replication and pathogenesis. Subcellular localization of the 39 candidate metal-associated MPXV proteins by Virus-mPLoc revealed that the majority (23 proteins, 56%) were localized in the host cytoplasm, highlighting

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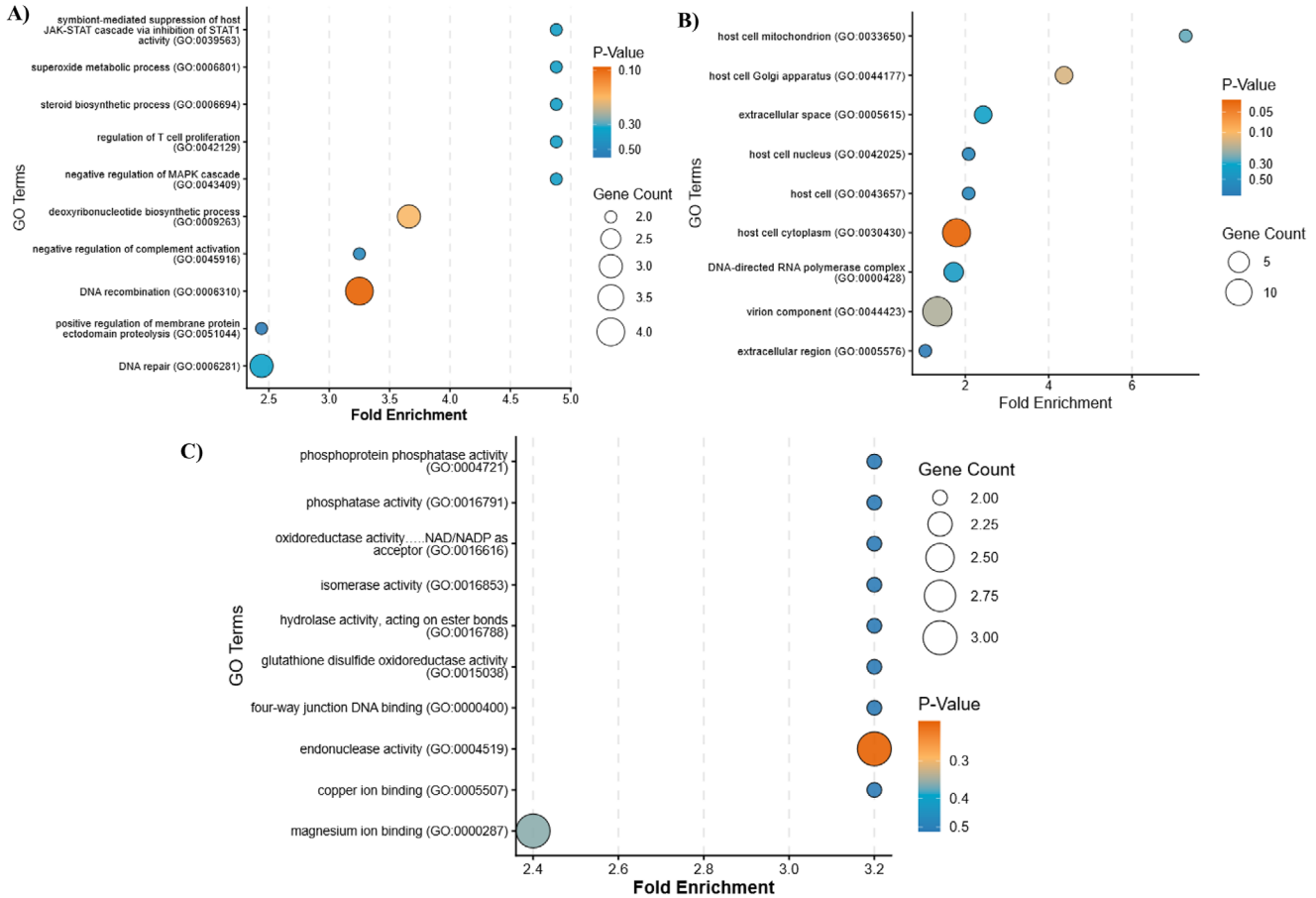


FIGURE 3 | Functional enrichment analysis of the MPXV metalloproteome. Bubble plots show the top-enriched Gene Ontology (GO) terms for 39 candidate metal-associated MPXV proteins across (A) biological process, (B) cellular components, and (C) molecular function. Enrichment highlights include DNA recombination (GO:0006310) and immune modulation (GO:0052170) in BP, viral envelope (GO:0044423) and replication complexes in host cell cytoplasm (GO:0030430) in CC, and metal-ion binding (GO:0046872), zinc-finger coordination (GO:0008270) and nucleases activities (GO:0004518) in MF. X-axis represents Fold enrichment; bubble size corresponds to gene count; bubble color indicates p-value. Analysis performed via DAVID and visualized using ggplot2.

the cytoplasm as the primary site for metal-dependent viral processes, followed by localization in the host nucleus, with 9 proteins (22%), suggesting their role in viral transcriptional regulation and nucleic acid metabolism. Only a small proportion (5 proteins, 12%) of these metal-associated proteins were predicted as "secreted", which suggest its possible participation in host-virus interaction and immune modulation. The remaining 4 proteins (10%) were predicted to be localized to the "host cell membrane", suggesting possible involvement in viral attachment, entry, assembly, or other host signaling pathways (Figure 5A).

The constructed visual network further illustrates the proposed relationship between, MPXV metal-associated proteins, their predicted subcellular localization within host cells and their affinities across the six metal ions (zinc, magnesium, manganese, iron, copper and cobalt). This diagram highlights that zinc- and magnesium-associated proteins constitute the most dominant fractions among the MPXV metalloproteome. With their major localization in the host cytoplasm (56%) and nucleus (22%), followed by the host cell membrane (10%). A smaller subset of these proteins is also predicted to be secreted (12%), suggesting their potential spatial roles in host-virus interactions (Figure 5B).

2.5 | Integrated Clustergram and Host-Virus PPI Network Analysis

The constructed phylogenetic tree was visualized as a clustergram, which revealed a clear segregation of MPXV metal-associated proteins into multiple distinct clusters. This tree was labelled from the cent to the periphery as subcellular localization, gene names, protein families, and metal-binding affinities (Zn, Mg, Mn, Fe, Cu, and Co) of MPXV metal-associated proteins. The outermost ring represents the corresponding UniProt accession IDs for each Metal-associated protein. The majority of these proteins are found to be associated with multiple metal ions. However, some of the proteins show affinity only for a single type of metal ion. Those single-metal-ion-associated proteins are highlighted as distinct branches in the network. (Figure 6).

The sequence alignment of 219 proteins of Vaccinia virus (UP000000344) with the 39 metal-associated proteins of MPXV using standalone BLASTp (e-value of 1×10^{-5}), followed by CD-HIT (at 90%) to remove any redundancy resulted in a total of 45 VACV protein sequences, showing high sequence similarity ($>$ 90%) to the corresponding MPXV metal-associated proteins.

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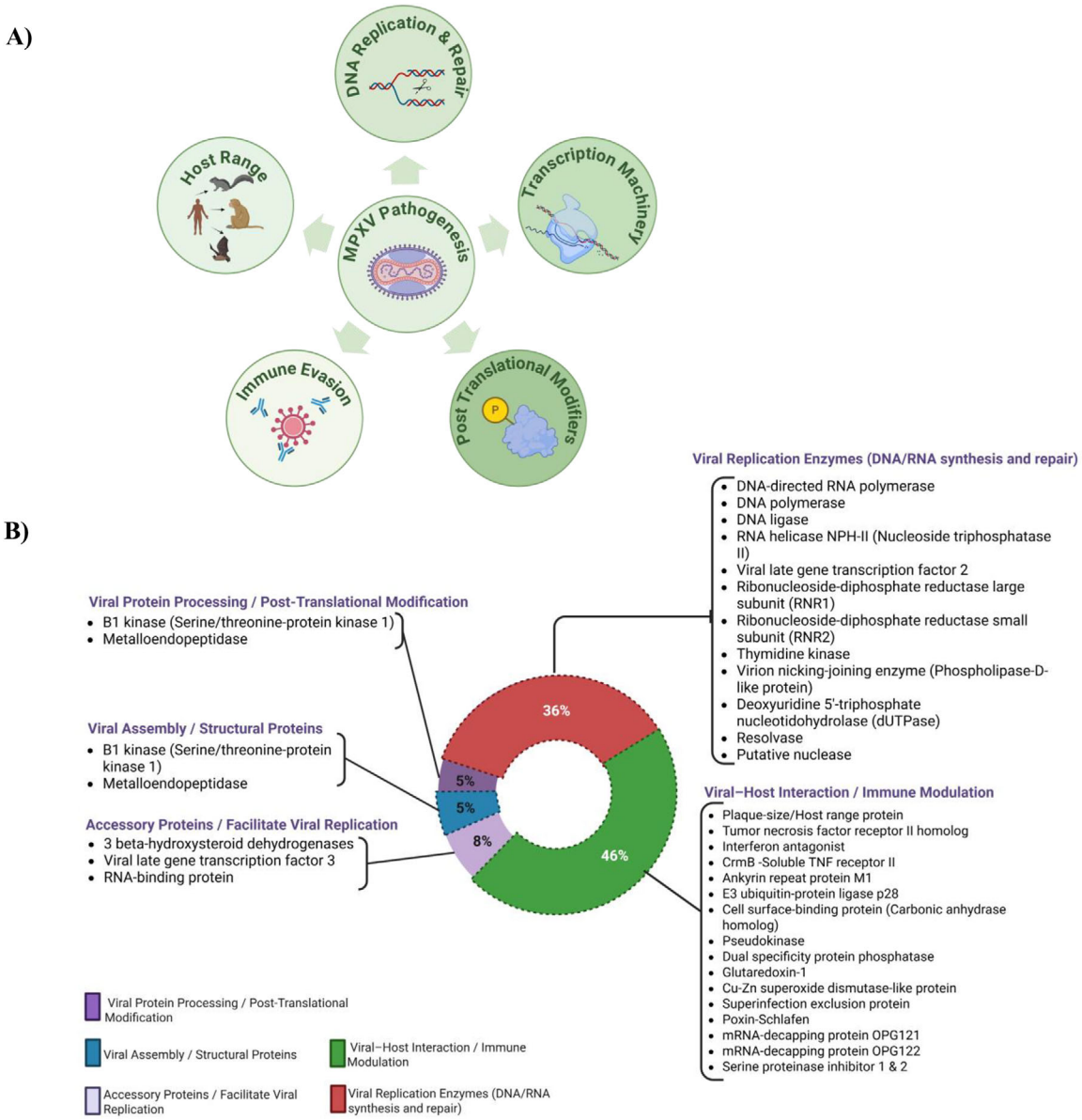


FIGURE 4 | Proposed functional categorization of candidate metal-associated MPXV proteins. (A) Major functional processes predicted to be involved in MPXV pathogenesis based on computational annotation, including DNA replication, transcription, protein processing, immune evasion, and host-range determination. (B) Distribution of the 39-candidate metal-associated proteins by predicted functional category: viral-host interaction (46%), DNA/RNA synthesis and repair (36%), accessory proteins (8%), protein processing and post-translational modification (5%), and viral assembly/structural proteins (5%). Assignments are based on InterProScan annotation and Gene Ontology analysis. These are computational predictions requiring experimental validation. Detailed annotations are provided in Table S2.

These 45 VACV proteins were queried against the human-virus protein-protein interaction dataset available in VirHostNet 3.0 for downstream network analysis. The edges and nodes were visualized in Cystoscope v3.10.4 as shown in Figure 7.

2.6 | Structural Validation by Performing PDB Homology Search and Alignment

To address the inherent limitations of computational structure prediction and to validate AlphaFold3 confidence matrix with experimental structure evidence, we systematically compared AF3predicted structure of candidate metal-associated MPXV proteins against experimentally determined homologous structures in the Protein Data Bank (PDB). Protein sequences were queried against PDB using BLAST-p, and PDB homologs meeting selection criteria of percentage identity $\geq 30\%$ and query coverage $\geq 70\%$ were retrieved. The BLAST-p outcome for all 39-candidate metal-associated MPXV proteins was tabulated. The MPXV metal-associated proteins were divided into five major categories: Category I, which belongs to the top seven of the 39-candidate metal-associated MPXV proteins (A0A7H0DN21, A0A7H0DN44, A0A7H0DNC3, A0A7H0DN73, A0A7H0DN15, A0A7H0DN77, A0A7H0DNC1). They are not only their PDB homologs that surpassed the threshold of $\geq$30% percentage identity and $\geq$70% query coverage, but they also have the same interacting metal as in pre-

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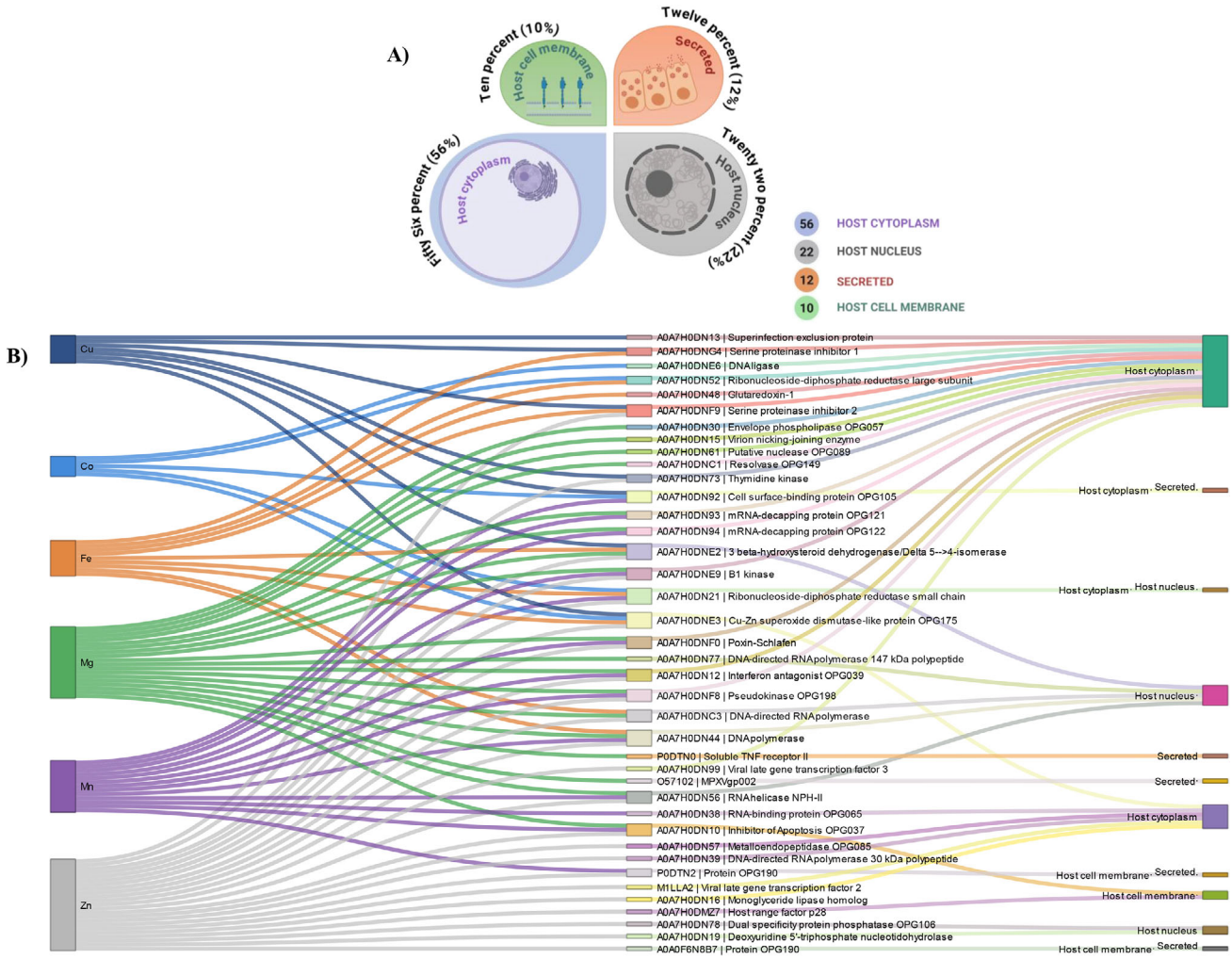


FIGURE 5 | Subcellular localization and function of potential metal-associated MPXV proteins in host cells. (A) The diagram depicts the percentage distribution of the metal-associated MPXV proteins across various cellular compartments, showing most enrichment in the cytoplasm (56%), followed by nucleus (22%), cell membrane (10%), and some as secreted proteins (12%). (B) Visual network of 39 candidate metal-associated proteins in MPXV across the six metal ions (Zn, Mg, Fe, Mn, Cu, and Co), The figure showcases MPXV protein identifiers (UniProt ID) and names, and link them to their predicted subcellular localization within the host cells. Width of the link represent the relative proportion of proteins associated with each metal type and subcellular compartment, highlighting the predominance of Zn and Mg binding proteins localized mainly to the host cytoplasm, nucleus, and cell membrane, with a subset being secreted. (Table S2).

dicted structure. Category II belongs to the six metal-associated MPXV proteins (A0A7H0DN61, A0A7H0DN94, A0A7H0DMZ7, A0A7H0DN57, A0A7H0DN99, M1LLA2). These proteins showed "no significant similarity found" in BLAST-p results. Category-III, these three proteins (A0A7H0DNE3, A0A7H0DNE2, O57102) were screened out since they do not fulfil the minimum threshold of $\geq$ 30% percentage identity and $\geq$70% query coverage. Category IV, where this one protein (A0A7H0DN52), though it surpassed the threshold, but its PDB homolog does not interact with the same metal as predicted by AF3. And last Category V, these proteins check the minimum threshold ($\geq$30% percentage identity and $\geq$70% query coverage), but its PDB homolog does not interact with any metal as per the experimental method through which this structure was retrieved as available in PDB database (details given in Supporting information S3).

Structure alignment using TM-align was performed on the candidate metal-associated MPXV proteins (A0A7H0DN21, A0A7H0DN44, A0A7H0DNC3, A0A7H0DN73, A0A7H0DN15, A0A7H0DN77, A0A7H0DNC1) that belong to the Category-I (Table 1).

Structural similarity was evaluated on the basis of TM-score metrics along with other crucial alignment parameters like aligned length, RMSD (root mean square deviation), sequence identity etc. Metal-coordinating residues were compared between AF3-predicted MPXV metal-associated protein and its PDB homolog derived from experimental validation. Based on the degree of structural and functional conservation, this multi-tiered validation approach stratified the seven-candidate metal-associated MPXV proteins as follows:

2.7 | High-Confidence Structurally Validated Candidates

Two candidate proteins (A0A7H0DNC3, A0A7H0DN73) achieved exceptional structural validation through perfect

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TABLE 1 | Structural validation metrics of seven high-confidence candidate metal-associated MPXV proteins with their experimentally derived PDB homolog.

MPXV Protein ID	Predicted metals	ipTM score (>0.8)	PTM score (>0.5)	PDB homolog ID	Interacting metal	Organism	Query Coverage ≥70%	Percent Identity ≥30%	MPXV A.A residues	PDB A.A residues	TM-Score	Aligned length	RMSD	Seq_N1ID	Confidence level	Experimental Method	References
A0A7H0DN21	Zn, Mn, Fe, Co	**0.9**	**0.83**	IHON	Co	*Mus musculus*	100	80.94	VAL73, GLN97, GLU101, HIS104, GLU197, PHE167, HIS200	GLU170, HIS173, ASP 139, ILE 263, HIS 270, GLU 267, GLU 233	0.87	282	0.74	0.74	High	X-Ray diffraction	[65]
A0A7H0DN44	Zn, Mg, Mn, Fe	0.88	0.85	8HG1	Mg	MPXV	100	99.9	ASP462, PHE262, ASP166, ILE167, ASP268	ASP753, ASP549, ILE789, GLU 790, GLU 792, ILE547, PHE548, SER754, CYS788, ASN551, TYR550	0.84	943	4.19	0.924	High	Electron microscopy	[66]
A0A7H0DNC3	Zn, Mg, Fe	0.88	0.92	6RFL	Zn, Mg	VACV	100	98.97	MG- NO INTERACTION, ZN- ASN1089, CYS1090, PRO1113, CYS1106, CYS 1103, CYS 1087, ARG1105	MG- ASN226, ASP415, ASP417, GLY 418. ZN- ASN1089, CYS1090, PRO1113, CYS1106, CYS1103, CYS1090, CYS 1087	0.96	1129	0.46	0.989	High	Electron microscopy	[24]

(Continues)

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TABLE 1 | (Continued)

MPXV Protein ID	Predicted metals	ipTM score (>0.8)	PTM score (>0.5)	PDB homolog ID	Interacting metal	Organism	Query Coverage ≥70%	Percent Identity ≥30%	MPXV A.A residues	PDB A.A residues	TM-Score	Aligned length	RMSD	Seq_N1ID	Confidence level	Experimental Method	References
A0A7H0DN73	Zn, Cu	0.97	0.86	9K6K	Zn	MPXV	100	99.44	CYS173, CYS170, LYS 140, CYS 141, CYS 138, PHE 142	CYS173, CYS170, LYS 140, CYS 141, CYS 138, VAL169	0.82	149	1.07	0.987	High	X-Ray diffraction	[23]
A0A7H0DN77	Mg	0.97	0.92	6RFL	Mg, Zn	VACV	100	99.07	ASP415, ASP417, ASP419, ARG 380,	ASP415, ASP417, ASP419, GLY418, ASN 226	0.2	393	9.09	0.051	Low	Electron microscopy	[24]
A0A7H0DNC1	Mg	0.94	0.78	5E6F	Mg	Canary pox	79	44.3	GLU101, GLU197, GLN97, HIS104, PHE167, HIS200, VAL73	GLU267, GLU233, GLU170, HIS 173, ASP139, ILE263	0.91	130	1.03	0.485	Moderate-High	X-Ray diffraction	[67]

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TABLE 1 | (Continued)

MPXV Protein ID	Predicted metals	ipTM score (>0.8)	PTM score (>0.5)	PDB homolog ID	Interacting metal	Organism	Query Coverage ≥70%	Percent Identity ≥30%	MPXV A.A residues	PDB A.A residues	TM-Score	Aligned length	RMSD	Seq_N1ID	Confidence level	Experimental Method	References
A0A7H0DN39	Zn	0.92	0.48	6RIC	Mg, Zn	VACV	100	97.68	SER164, CYS187, CYS190, ASN161, CYS159, CYS162, LYS159	ZN3(1302)-ALA43, GLY42, CYS59, GLY61, THR51, CYS62, GLY53, CYS49, HIS62.ZN 8 (1201)-ASP1092, CYS1090, ASN1089, CYS1087, CYS1106, CYS1103. ZN 2 (101)-CYS34, CYS40, ASN37, THR9, CYS7, GLY11	0.33	146	5.87	0.068	Low	Electron microscopy	[68]

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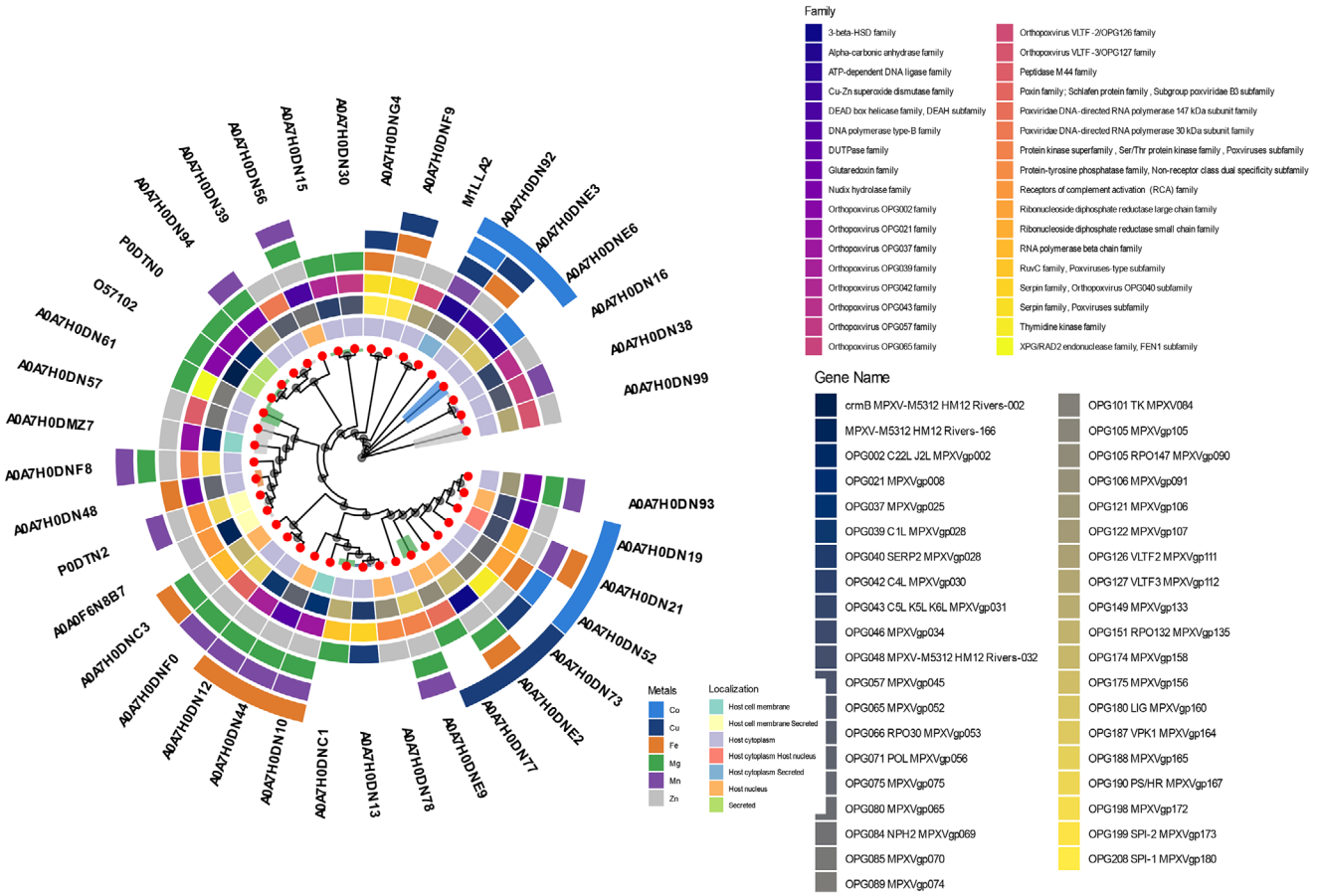


FIGURE 6 | Circular clustergram of candidate metal-associated proteins of Monkeypox virus (MPXV). (The concentric rings depict, from the center to the periphery, subcellular localization, gene names, protein families, and metal-binding affinities (Zn, Mg, Mn, Fe, Cu, and Co) of MPXV metal-associated proteins. The outermost ring represents the corresponding UniProt accession IDs for each MPXV protein. The network of metal-binding proteins in Monkeypox virus (MPXV). The highlighted branch on the tree shows the single metal ion binding affinity of that protein.

conservation of metal-coordinating residues. A0A7H0DNC3-DNA-directed RNA polymerase (Zn & Mg associated) demonstrated identical coordination geometry across all metal binding sites. The Zn-coordinating residues predicted by AF3 (ASN1089, CYS1090, PRO1113, CYS1106, CYS1103, and CYS1087) matched exactly with experimentally resolved coordinates in the Vaccinia virus homolog (PDB 6RFL). Structural superimposition yielded exceptional alignment metrics: TM-Score 0.96, RMSD 0.46 Å and 100% sequence identity across the aligned region (98.97% overall identity, 100 percent query coverage). This represents the highest confidence structural validation in our study, demonstrating that AF3 accurately predicted the precise three-dimensional metal-binding pocket geometry (Figure 8A).

In A0A7H0DN73 (Thymidine Kinase), a zinc-associated MPXV protein, all coordinating cysteine residues (CYS173, CYS170, CYS141, CYS138) showed perfect positional conservation with identical positions in the MPXV PDB homolog (PDB 9K6K). TM-score of 0.82 and RMSD of 1.07 Å along with high sequence identity (99.44%) and 100% query coverage, indicates excellent structural alignment (Figure 8B).

2.8 | Candidates Showing Functional Conservation With Positional Variation

MPXV metal-associated proteins (A0A7H0DN44, A0A7H0DNC1, A0A7H0DN21) demonstrated robust functional conservation of metal coordinating residuals despite positional variation. Superimposition between A0A7H0DN44 (Mg-associated) and MPXV PDB homolog (PDB 8HG1), retained the same coordinating residues type (ASP, GLU, ILE), though with positional shifts of ±3 to 240 amino acid positions due to large-scale domain rearrangements. Despite this architectural variation, TM-score of 0.84 and RMSD of 4.19 Å confirmed overall structural homology, and the functional conservation of acetic acid residues that may be crucial for magnesium coordination.

A0A7H0DNC1 (Mg-associated) showed conservation of Mg-coordinating residue types (GLU, HIS, ASP)) between AF3 and Canary pox homolog (PDB 5E6F), classified as moderate-high confidence. Despite lower sequence identity (43%), the TM-Score 0.91 and RMSD 1.03 Å indicated reliable structure alignment and the presence of conserved acidic/histidine residue suggest Mg-coordination capability.

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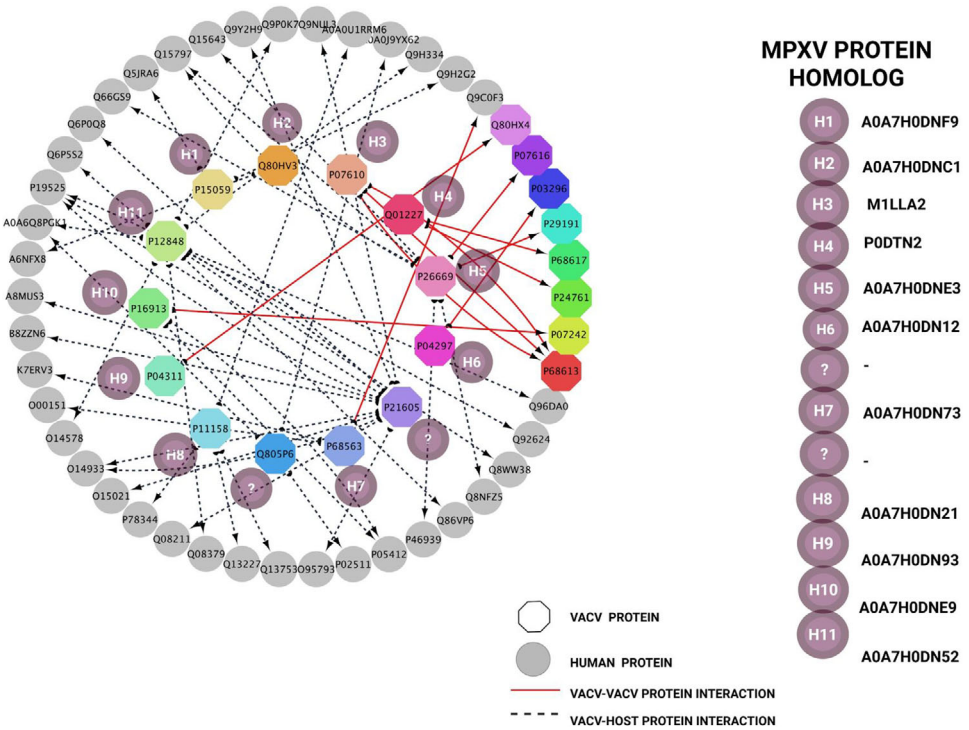


FIGURE 7 | Human host-vaccinia virus protein-protein interaction (PPI) network highlighting MPXV homologs. The network illustrates protein-protein interactions among Vaccinia virus (VACV) proteins, their Monkeypox virus (MPXV) homologs, and human host proteins. Colored nodes represent MPXV homologous proteins identified by sequence similarity and functional correspondence with VACV proteins. VACV proteins are shown as colored nodes within the viral interaction core, while human host proteins as grey nodes. The network comprises 59 nodes, including viral and host proteins, connected by 54 edges, reflecting both intra-viral and host -virus interactions.

AlphaFold3 predicted A0A7H0DN21 (Co-associated) structure, superimposed with *Mus musculus* homolog (PDB IHON) demonstrated functional conservation of metal-associated residues (HIS, ASP, GLU, CYS) with moderate positional flexibility, supported by a TM-score of 0.87 and RMSD 0.74Å.

2.9 | Complex Metal Coordination: Multi-Site Binding

A0A7H0DN39 (Zn-associated) presented a one-of-a-kind case where the experimentally derived PDB homolog (PDB 6RIC) resolved 3 distinct Zn-binding sites with non-identical coordinating residue sets with slight overlap. On the other hand, AF3 predicted the primary Zn-binding side with coordinating residues (SER164, CYS187, CYS190, ASN161, CYS159, CYS162), which partially matched one of the three experimental sites (ZN3:ALA43, GLY42, CYS59, GLY61, THR51, CYS62, GLY53, CYS49, HIS62). The low TM-score (0.33) and RMSD (5.87 Å) reflect the complexity of a multi-site coordination, suggesting that this protein may employ multiple metal-binding mechanisms.

2.10 | Low Structural Alignment Despite High AF3 Confidence

A0A7H0DN77 (Mg-associated) protein with a AF3 confidence metrics (ipTM 0.97, pTM 0.92) suggested reliable structure prediction, yet structural superposition with the VACV homolog (PDB 6RFL) yielded a very low TM-score (0.2) with high RMSD (9.09 Å), indicating random structural similarity. However, the presence of ASP residues (ASP415, ASP417, ASP419) in both AF3 and experimentally derived PDB structure suggests functional conservation despite poor overall fold alignment. This case illustrate that AF3 confidence metrics alone cannot fully validate metal-binding prediction without structural comparison.

In total, 7 proteins (A0A7H0DN21, A0A7H0DN44, A0A7H0DNC3, A0A7H0DN73, A0A7H0DN39, A0A7H0DN77, A0A7H0DNC1) (17.9%) achieved structural validation supporting potential metal-binding predictions, while 32 proteins failed to qualify for downstream analysis as with 22 proteins (A0A7H0DNE6, A0A7H0DN92, A0A7H0DNE9, A0A7H0DN48, A0A7H0DNF9, A0A7H0DNG4, A0A7H0DN13, A0A7H0DN30, A0A7H0DN56, A0A7H0DNF0, P0DTN0, A0A7H0DN10, A0A7H0DN12, A0A7H0DN93, A0A7H0DNF8, P0DTN2, A0A7H0DN38, A0A7H0DN78, A0A0F6N8B7, A0A7H0DN19, A0A7H0DN16, A0A7H0DN15) with moderate to good structure homology to their PDB homolog, decent alignment scores, but lacked information regarding any potential metal-binding site conservation in available PDB structures as per the experimental method on record and for the rest 10 MPXV proteins, they either fail to surpass the minimum threshold (≥30% percentage identity and ≥70% query coverage) or "no significant similarity found" in BLASTp results against Protein Data Bank or lack common metal

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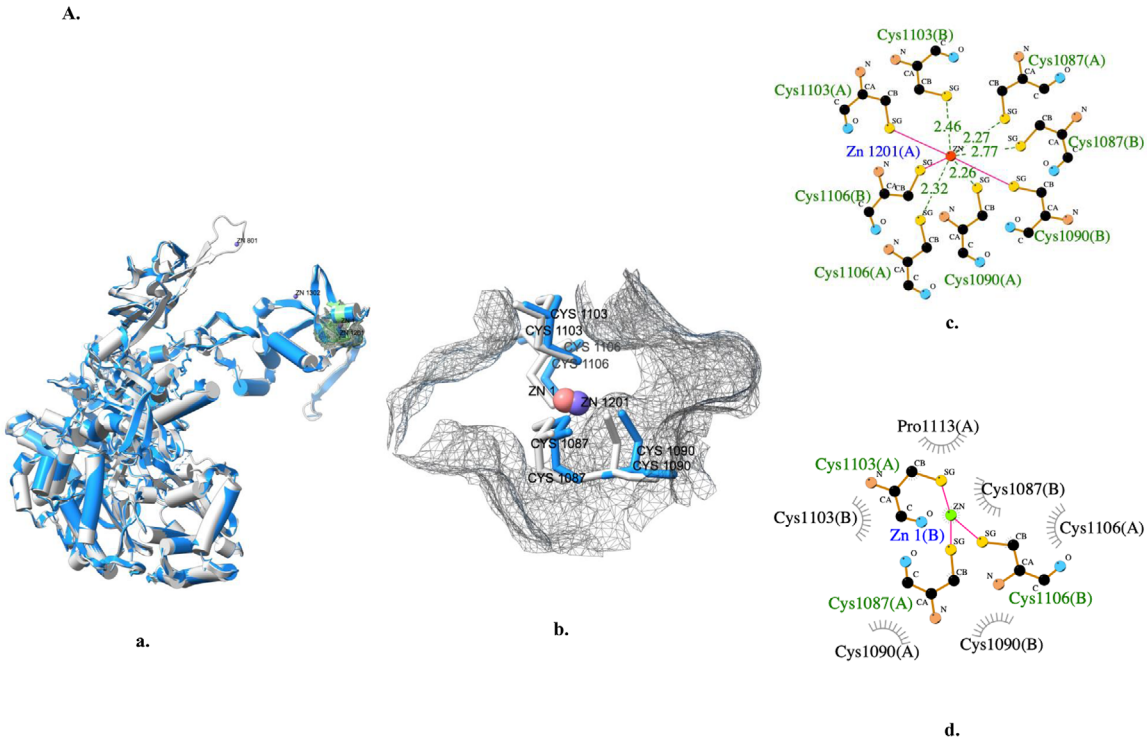


FIGURE 8A | (a) Structural superimposition of AF3 predicted MPXV DNA -directed RNA polymerase A0A7H0DNC3 (Chain B, grey) with PDB homolog 6RFL of VACV (Chain A, blue) (TM-score 0.96, RMSD 0.46Å). The metal-binding zone within a 5 Å radius is demarcated. (b) Zinc-binding site (Zn 1201) embedded within the 5 Å focal area electron density mesh, coordinated by CYS1103, CYS1106, CYS1090, CYS1087) with electron density mesh. (c) Two-dimensional ligand interaction diagram between Zn 1201 and metal-binding residues within the metalloprotein structure. (d) Two-dimensional ligand interaction diagram of secondary zinc-binding site Zn 1(B) and metal-binding residues within the metalloprotein structure.

between the prediction and experimentally derived structure, as elaborated in Supporting information S3.

3 | Discussion

Monkeypox virus (MPXV), an emerging zoonotic virus with a double stranded linear DNA, belongs to the genus Orthopoxvirus. This enveloped virus has a genome of approximately 197 kb, capable of coding for 183 proteins. MPXV has traditionally been studied through the lens of comparative genomics and structural homology with its close relative Vaccinia virus, which also belongs to the Poxviridae family [12].

The functional analysis of viral proteomes remains a cornerstone of understanding pathogenesis and developing antiviral strategies. Metal-associated proteins play crucial role throughout the life cycle of the virus, including genome replication, transcription, protein structure stabilization, and evasion of host immune responses [5, 6]. However, the exact role of metal ions in MPXV biology has remained largely unexplored. In this study, we propose a comprehensive proteome-scale pipeline in which we performed a homology-based, structure-guided characterization of the MPXV metalloproteome, identifying 39 candidate metal-associated proteins that may interact with transient metals, including zinc, magnesium, manganese, iron, copper, and cobalt. These metals perform a myriad roles inside the viruses such as, acting as metal cofactors contributing to the structural integrity (zinc finger motifs) and enzymatic catalysis (polymerases and kinase) [13, 14].

Employing a metal-centric computational investigation, integrating a sequential framework of local alignment of protein sequences, followed by structural prediction with AF3. We established a robust in identifying these candidate metal-associated proteins within the MPXV proteome [15–18].

Resulted in a total candidate set to 39 potential, distinct, high-confidence metal-associated proteins. This curated and proposed metalloproteome remains predictive in nature and is dominated by zinc (31%), magnesium (23%), and manganese (17%) binding proteins, while iron (13%), copper (9%), and cobalt (7%) respectively. Though they constitute a smaller functional subset, yet it substantially expands existing annotation database. Notably, 19 of these proteins demonstrated predicted affinity for distinct and multiple metal ions. Showcasing a diverse spectrum of metal-associations within the MPXV metalloproteome.

The functional analysis of the putative metal-associated protein, using DAVID-based GO term enrichment and KEGG Orthology (KO) assignment, we proposed that the MPXV metalloproteome forms an integrated functional network, orchestrating the viral life cycle [19–21].

The most relevant biological enrichment was observed in DNA recombination (GO:0006310) and deoxyribonucleotide biosynthesis (GO:0009263), biologically supported by the RNR complex (Ribonucleoside-diphosphate reductase complex), comprising two subunits, the large subunit (A0A7H0DN52/K10807) and small subunit (A0A7H0DN21/K10808). In Poxviruses, it

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is already established that the RNR complex is an iron-dependent metalloenzyme that catalyzes the rate-limiting step of dNTP production [22]. Our mapping to purine (map00230) and Pyrimidine (map00240) metabolism hierarchies strengthens the idea that MPXV may maintain an autonomous, metal-dependent pathway for genome replication and expansion. While RNR is a known poxvirus feature, our analysis places it as a primary node within a broader metal-coordinated network that includes the magnesium-dependent Thymidine kinase (A0A7H0DN73/K00857), suggesting a highly prioritized metal demand for maintaining the dNTP pool balance during rapid cytoplasmic replication [23].

The enrichment of the host cytoplasm (GO:0030430) and virion component (GO:0044423) suggests a strategic spatial distribution of these proteins, especially the localization of DNA-directed RNA polymerase (A0A7H0DN77/K25273) within the virion component, which is particularly significant. Mapping to the virion-Orthopoxvirus (03268), this hierarchy suggests that MPXV prepackages this magnesium-dependent machinery to autonomously initiate an immediate early gene expression upon entry [24, 25]. This "pre-loading" strategy, combined with cytoplasmic metabolic enzymes such as the zinc-dependent monoglyceride lipase (A0A7H0DN16/K01054), implies that MPXV creates a metal-enriched microenvironment within the host cytoplasm. The enrichment of this viral protein cellular component suggest that majority of the metal-dependent structural and enzymatic processes are localized to function within the "viral factories". Ensuring to maximize the efficiency of Virogenesis, a natural phenomenon well studied in Vaccinia virus morphogenesis [26–29].

For the GO:MF enrichment, the most significant terms include "metal ion binding" (GO:0046872), and "zinc ion binding" (GO:0008270). Which further support the earlier functional annotation result indicated by Cu-Zn superoxide-dismutase (A0A7H0DN15/K16860) and carbonic anhydrase (A0A7H0DN92/K01672). Earlier known in managing host-induced oxidative stress and maintaining pH homeostasis, respectively [30, 31]. Also, the presence of zinc finger domain and catalytic motif domain suggest that these viral metal-associated proteins are structurally conserved and evolutionarily constrained [32–34].

High enrichment of these metal-associated proteins within the host cytoplasm to form viral factories suggest synchronized metal dependent mechanism that relies heavily on the metal cofactors at the site of replication and assembly. A similar trend was also observed in the VACV infectious pathway where the recruitment of zinc-finger domain protein such as, G4L plays a critical role in viral morphogenesis and redox regulation [34]. With reference to that, it suggest that in MPXV by regulating the metal ions in their respective intracellular locations within the host cell can accounted as a prom could offer a promising targeted antiviral approach.

With the current lack in availability of experimental evidences on human-MPXV protein-protein interaction (PPI). We tried to identify the metal-associated protein orthologs within the VACV proteome. This resulted in 45 VACV hits, which was further analyzed for virus-host interaction VirHostNet 3.0 database. Revealing a direct PPI association with 21 human host proteins, which further resolved to 11 metal-associated proteins of MPXV which are homologs of Vaccinia proteins, visualized in Cytoscape v3.10.4 [35–37]. Particularly those linked to immune modulation (H 1- A0A7H0DNF9, H4-P0DTN2, H5-A0A7H0DNE3, H6-A0A7H0DN12, H9-A0A7H0DN93), viral replication (H2-A0A7H0DNC1, H7-A0A7H0DN73, H8-A0A7H0DN21, H11-A0A7H0DN52), transcriptional regulation (H3-MILLA2), and virion morphogenesis (H10-A0A7H0DNE9) (Figure 7). This result emphasizes the evolutionary conservation of these metal-associated proteins across Orthopoxviruses. Making Vaccinia an ideal substitute model for inferring the dynamics of MPXV-host interactome.

One of the critical limitations of the above computational approaches to predict the metal-associated proteins is the potential for false positives, where sequence similarity to metal-binding proteins alone does not guarantee the proposed metal-binding capability or functional relevance, as it lacks support from direct site-level structural evidence. To address this concern, we performed systematic structural validation by aligning AF3 predicted metal-associated MPXV protein structures against experimentally derived PDB homologs from Protein Data Bank by performing BLAST-p (threshold of ≥ 30% percentage identity, ≥ 70% query coverage). This structural superimposition with solved metal-bound homologs revealed important insights into prediction accuracy and conservation of known coordinating residues.

Perfect positional conservation of metal-coordinating residues in the two metal-associated MPXV proteins A0A7H0DNC3 (Mg & Zn-associated DNA-directed RNA polymerase with 6RFL) and A0A7H0DN73(Zn-associated Thymidine kinase with 9K6K) provides strong and compelling evidence in support of AF3 prediction accuracy in Monkeypox virus (Figures 8A and 8B).

Alignment metric score of TM-score from 0.82-0.96 and RMSD <1.1Å shows the accuracy of computational 3D structure prediction of geometry required for metal binding. It also confirms the geometry for metal selectivity, coordination strength and catalytic activity [38]. Detailed alignment metrics are documented in Table 1.

Functional conservation of coordinating amino acid residues (ASP, GLU, HIS & CYS) in A0A7H0DN44 (Mg-associated DNA polymerase with 8HG1), A0A7H0DNC1 (Mg-associated Resolvase with 5E6F), AOA7HODN21 (Co-associated Ribonucleoside-diphosphate reductase small chain with IH0N), and A0A7HODN39 (Zn-associated DNA-directed RNA polymerase with 6RIC) and a moderate-high TM-score (0.33-0.97) suggest that MPXV metalloproteome may have a flexible metal-binding strategy which imparts evolutionary advantage, with viral adaptation or host-range expansion [39–41]. Consistent coordination of residue types supports the association of zinc, magnesium, and cobalt with these MPXV proteins. The alignment of 6RIC with A0A7HODN39 (Zn-associated DNA-directed RNA polymerase) also exhibited a unique scenario where the experimentally derived PDB homolog (6RIC) resolved 3 distinct Zn-binding sites, with few non-identical overlapping coordinating residue sets, while the AF3 predicted the zinc-association with the coordinating residues (SER164, CYS187,

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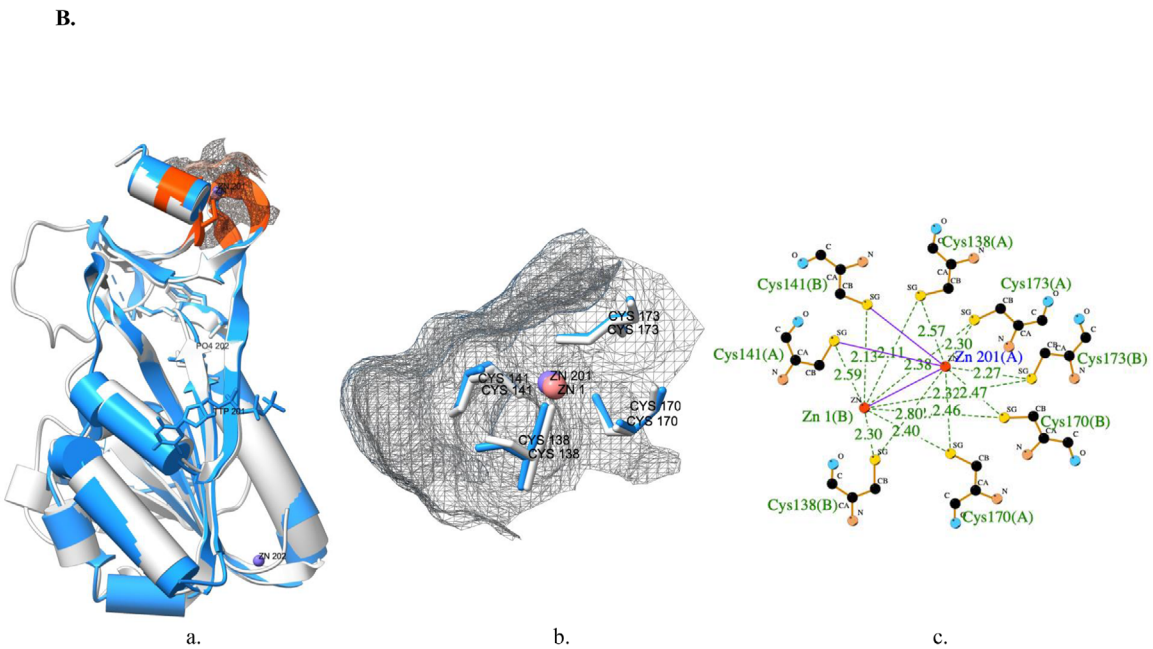


FIGURE 8B | (a) Structural superimposition of AF3 predicted MPXV Thymidine Kinase A0A7H0DN73 (Chain B, grey) with PDB homolog 9K6K of MPXV (Chain A, blue) (TM-score 0.82, RMSD 1.07Å). The metal-binding zone within a 5 Å radius is demarcated. (b) Zinc-binding site (5 Å focal area) coordinated by CYS138, CYS141, CYS170, and CYS173 with electron density mesh. (c) Two-dimensional ligand interaction diagram between Zn 1 (B) and Zn 201 (A) and metal binding residues within the metalloprotein structure.

CYS190, ASN161, CYS159, CYS 162), which matched partially with one of the three experimental sites (ZN3:ALA43, GLY42, CYS59, GLY61, THR51, CYS62, GLY53, CYS49, HIS 62) also documented in Table 1. The low TM-score (0.33) and RMSD (5.87) with multi-site coordination hint that the protein may use a multi-metal-binding mechanism as an evolutionary adaptation in maintaining metal homeostasis and enzymatic flexibility [42, 43].

The structural superimposition of A0A7H0DN77 (Zn- and Mg-associated MPXV protein) with its closest PDB homolog 6RFL demonstrated poor alignment with very low TM-score (0.2) and a high RMSD (9.09 Å), indicating a random structural similarity in spite of high AF3confidence metrics (ipTM0.97, pTM0.92), further reiterating that AlphaFold3 confidence metrics alone cannot fully validate metal-binding prediction without structural alignment.

In summary, seven MPXV proteins (17.9%) achieved structural validation, supported by confident alignment metrics, with varying degrees of functional and positional conservation between the AF3-predicted, metal-associated MPXV proteins and their PDB homologs, as tabulated in Table 1.

The above computational and structural validation approaches revealed suggestive functions of the metal-associated MPXV proteins, and their strategic spatial organization was predicted using Virus-mPLoc server [44]. Based on our findings, we proposed that the MPXV metalloproteome may form a coordinated network encompassing five broad functional categories, including: (1) viral replication enzyme involved in DNA /RNA replication and repair, (2) viral-host interaction and immune modulation, (3) viral protein processing and post-translational modification, (4) viral assembly and structural protein, and 5. Accessory proteins facilitating viral replication. A working hypothesis of the predicted functional roles of candidate metal-associated MPXV protein across the viral life cycle is proposed in Figure 9.

This metalloproteome-centric view of MPXV infection provides new translational opportunities to combat pathogenesis. Viral metal-associated proteins represent attractive anti-viral targets, as their catalytic activity and structural integrity are often sensitive to metal availability [4, 45]. In SARS-CoV-2, for instance, zinc-ejecting agents like Disulfiram and Ebselen were shown to irreversibly destabilize the zinc-finger domains of Nsp 13 helicase and Nsp 14 exoribonuclease, effectively stripping the virus of its replicative fidelity [46].

Consequently, the repurposing of FDA-approved ionophores or metal chelator strategies has been demonstrated in coronaviruses, which may also provide a practical strategy to target MPXV using existing drugs, awaiting the development of new treatments [47]. Recent proteome-wide analyses of DNA viruses further support the feasibility of metalloproteome-guided antiviral discovery, highlighting conserved metal-binding hubs as druggable vulnerabilities [5]. Our systematic expansion of the MPXV metalloproteome thus lays a foundation for rational exploration of metal-targeted therapeutic strategies.

Our findings are predictive in nature, relying on computational homology modeling and sequence conservation analyses. No direct structural, biochemical, mutational, or metal content validation has been performed. Additionally, this analysis represents a static model, which does not capture dynamic fluctuations in intracellular metal pools during viral infection [48, 49].

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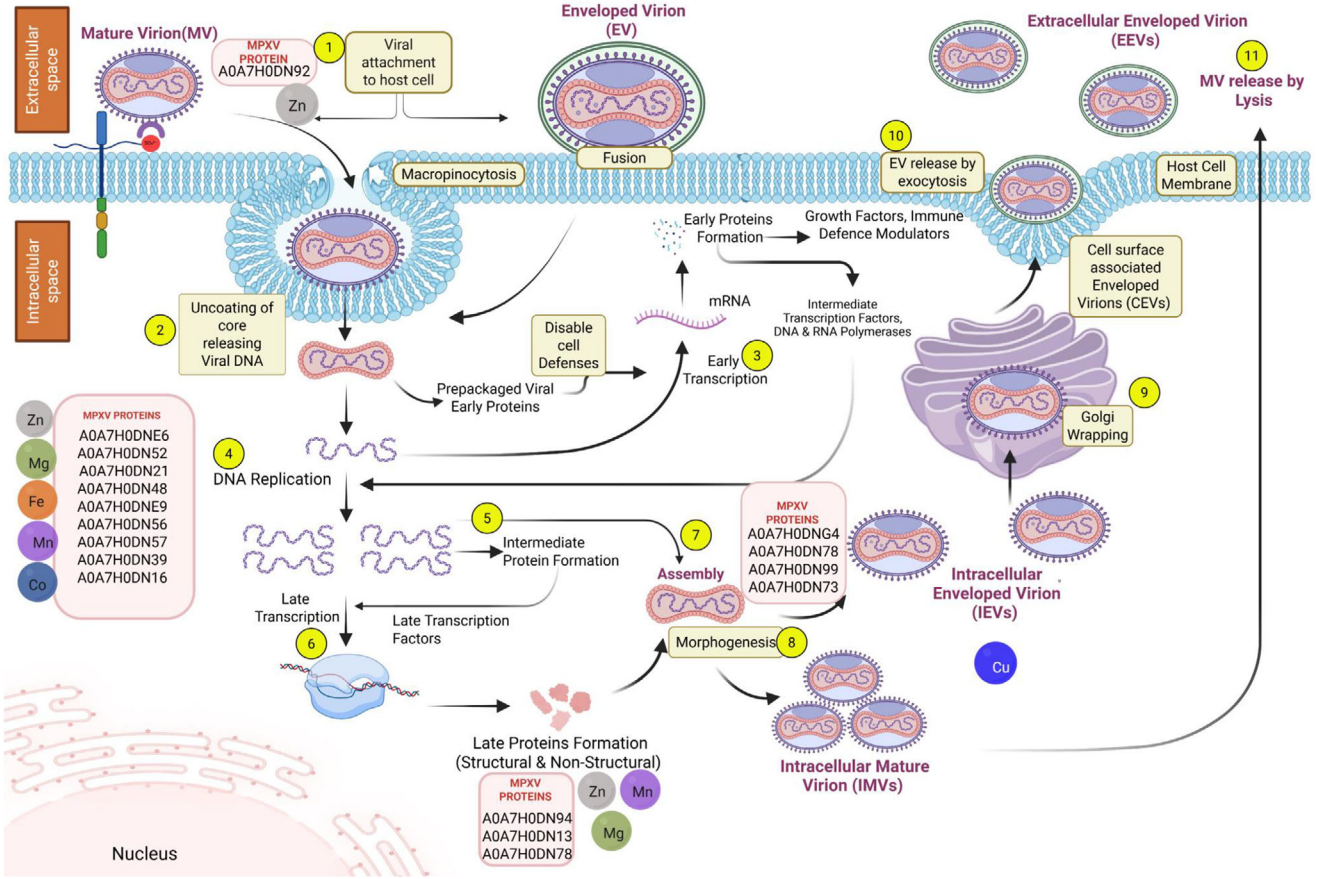


FIGURE 9 | Predicted functional roles of candidate metal-associated MPXV proteins across the viral life cycle (working hypothesis). A computational framework integrating functional annotation, subcellular localization prediction, and phylogenetic analysis to propose potential roles of candidate metal-associated MPXV proteins during infection. Based on InterProScan domain analysis and Gene Ontology enrichment, proteins are computationally assigned to putative infection stages (entry, early transcription, DNA replication, intermediate protein formation, late transcription, assembly, and release). Metal associations (Zn, Mg, Fe, Mn, Cu, and Co) are indicated with colored circles and are based on sequence homology and domain predictions. across multiple stages of infection. Zn-dependent proteins such as the Metalloendopeptidase (A0A7HODN57) mediate viral protein processing, while Fe and Co binding Ribonucleotide reductase RNR1 (A0A7H0DN52), Zn and Cu binding Thymidine kinase, and the Mgependent Resolvase (A0A7H0DNC1) support viral DNA synthesis and repair. MBPs involved in viral-host interaction, including the Zn-binding cell surface. Functional assignments are inferred from annotations alone and reflect computational hypotheses rather than experimentally validated roles. For example, Zn binding metalloproteases (A0A7HODN57) are predicted to participate in protein processing, Fe and Cobinding ribonucleotide reductase (A0A7H0DN52) may support DNA synthesis, and Zn-binding proteins (A0A7H0DN92, A0A7HODN78) are predicted to facilitate host interaction based on functional domain homology. MPXV proteins associated with assembly (A0A7H0DN30, A0A7H0DN16) are predicted to participate in virion morphogenesis based on localization and functional predictions. These assignments should be interpreted as working hypotheses requiring experimental validation through mutagenesis studies, a biochemical essay, and direct functional analysis. Detailed information on all candidate metal-associated proteins and their functional annotation is provided in Table S2.

Therefore, these results should be interpreted as a computational framework that provides hypothesis for future experimental validation. Subsequent studies employing site-directed mutagenesis, metal-depletion assays, and biophysical characterization (X-ray crystallography, IC-PMS-Inductively coupled plasma mass spectrometry) are warranted to validate predicted metal-coordinating residues, determine precise metal stoichiometry, and binding affinities of 39 candidate metal-associated MPXV proteins.

This study provides an extensive computational analysis of candidate metal-associated proteins in MPXV by integrating sequence homology search, structural prediction, and functional analysis. We identified 39 candidate MPXV proteins with predicted association to six different metals (zinc, magnesium, manganese, iron, copper, and cobalt) and were able to strengthen the prediction by performing computational structural validation, based on which we characterized their potential roles in viral replication, transcription, virion assembly, and host-virus interaction. Though our findings are entirely computational, they lay a systematic framework for prioritizing candidates for future experimental validation. Overall, the integration of structural, evolutionary, and functional analysis expands the known repertoire of candidate metal-associated MPXV proteins and proposes potential avenues for identifying conserved biochemical vulnerabilities in orthopoxviruses. As emerging orthopoxviruses like MPXV continue to pose public health threats, this metalloprotein-focused roadmap may guide and pave the way to the rational

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design of antiviral interventions through experimental studies.

4 | Material and Methods

4.1 | Retrieval of Monkeypox Virus (MPXV) Proteome

The reference proteome of the MPXV (UP000516359) was retrieved from the UniProt Proteome database (https://www.uniprot.org/proteomes/UP000516359); date of data retrieval: 4 June 2025. UniProtKB (UniProt Knowledgebase) is a comprehensive publicly accessible repository of protein sequences and their functional analysis. It integrates vast experimental data, computational prediction, and literature, which serves as a reliable and open source for protein identification and comparative analysis. The proteome database provides detailed information on the complete set of proteins expressed by an organism and hence facilitates organism-wide studies. In this study, we retrieved the complete reference proteome of MPXV (UP000516359) with a total of 183 proteins, which were downloaded as FASTA sequences for downstream sequence homology search and prediction of structure and functional analysis. These 183 proteins will now serve as a primary dataset to perform subsequent analysis. The overall workflow of this computational analysis is depicted in Figure 1.

4.2 | Compilation of Metal-Wise Reference Data Sets

To generate reference data sets for known metal-binding proteins, UniProtKB was queried individually for proteins annotated as metal-binding (date of retrieval: 17 May, 2025) [50, 51]. For this study, we have selected six metals (zinc, magnesium, manganese, iron, copper, and cobalt) across all domains of life based on their importance and relevance inthe life cycle of a virus and their pathogenesis as per the literature survey. The sequence search was done using the keywords "zinc binding protein" and "magnesium binding protein" on UniProt KB. The search was made on the basis of finding similar sequences or motifs characteristics for that particular metal. Metal-wise hits were retrieved and downloaded as FASTA sequences. These will serve as individual metal reference dataset for comprehensive sequence similarity analysis with the MPXV proteome (https://www.uniprot.org/uniprotkb?query=zinc+binding+protein).

4.3 | Sequence Similarity Search Using S tandalone BLASTp

A local environment was configured using the NCBI BLAST-2.15.0+ suite. The MPXV proteome served as the query set, while each of the six metal-binding protein datasets was used as a subject database. Pairwise sequence similarity searches were performed using BLASTp, applying an e-value cutoff of 1×10^{-4} and requiring percentage identity >50%. Analyses were accelerated by distributing the workload across 16 computational threads, and the results were exported in a tab-delimited format. This resulted to 722 hits for: zinc (466), magnesium (126), manganese (38), iron (44), copper (41), and cobalt (7) respectively.

4.4 | Data Refinement and Validation

To remove BLASTp redundancy, the duplicate entries were manually removed which resulted to 110 putative metal-associated proteins: zinc (35), magnesium (30), manganese (16), iron (15), copper (15), and cobalt (5) respectively. All the protein IDs (UniProt identifiers) were subjected for their protein sequence retrivaltheir corresponding FASTA files as presented in Supporting information S1.

4.5 | Three-Dimensional Structure Prediction Using AlphaFold 3

The 3D structures of 39 candidate metal-associated MPXV proteins were predicted using AlphaFold3 (AF3), an advanced deep-learning platform capable of predicting protein structure with high accuracy [15, 17, 18]. The predicted models were screened based on the ipTM and pTM confidence metrics (ipTM > 0.8 and pTM > 0.5). Where predicted template modelling score (pTM > 0.5) measures the overall accuracy of the predicted global tertiary structure and its measure of how similar the predicted structure might be to the real structure, while interface predicted template modelling score (ipTM > 0.8) measures how accurate is the relative positions of predicted subunits within the complex.

The candidate MPXV metal-associated proteins, who surpassed the confidence threshold (ipTM > 0.8 and pTM > 0.5), were shortlisted for downstream analysis and are documented in supplementary file S1. Though these confidence metrics do not directly validate the metal binding capability of the protein, they do serve as an initial screening mechanism which requires subsequent structural validation [15].

4.6 | Functional and GO Annotation

Functional annotation of the shortlisted metal-associated proteins was performed using the InterProScan v5.76-107.0, which integrates information from signature-recognition databases such as PFAM, CDD, GENE3D to provide comprehensive functional analysis and classification of proteins [52–54] InterProScan is a Java-based program, that perform functional analysis by scanning protein sequences against a large array of databases, utilizing position-specific scoring matrices and a hidden Markov model (HMM) which functionally classify proteins. InterProScan annotation results revealed conserved domains, protein families, and associated GO terms for identified MPXV metal-associated proteins. The functional annotations provide a framework to predict their roles across the viral life cycle. Evidence collated from multiple predictive models considerably enhances the confidence of prediction and functional assignment. The functional annotation results were tabulated, summarizing the proposed role of candidate metal-associated proteins in the life cycle of MPXV.

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To identify significantly enriched biological pathways for biological processes and molecular function, Gene Ontology enrichment analysis was performed on the 39 candidate metal-associated MPXV proteins. The analysis was performed on DAVID (Database for Annotation, Visualisation and Integrated Discovery) with a p-value threshold of 0.05 and multiple testing correction applied using Benjamini-Hochberg FDR [19, 55]. Ancestral charts of the top Gene Ontology terms for biological processes (BP), cellular components (CC), and molecular functions (MF) were also retrieved from QuickGO [56].

4.7 | KEGG Pathway Annotation and Ortholog Assignment

The KEGG pathway analysis was performed using the GhostKOALA server (https://www.kegg.jp/ghostkoala/) to identify metabolic and signaling pathways potentially associated with candidate metal-associated MPXV proteins. Protein sequences of the 39 candidate metal-associated MPXV proteins were submitted to GhostKOALA for Ortholog assignment using the viral reference databases [54]. Out of 39, 30 MPXV proteins were assigned successfully, and 9 remained unassigned. Successful ortholog assignments were mapped to KEGG orthology identifiers and integrated into KEGG pathway maps to identify enriched metabolic and signaling pathways relevant to viral replication and host interaction. Details of KO assignments are given in Supporting information S4.

4.8 | Subcellular Localization Prediction and Clustergram Construction of Predicted MPBs

Subcellular localization prediction provides insight into protein function and potential roles in viral replication and host-virus interaction. The Virus-mPLoc server was used to predict the subcellular localization of the 39-candidate metal-associated MPXV proteins. (http://www.csbio.sjtu.edu.cn/bioinf/virus-multi/) [44].

The Virus-mPLoc server integrates gene ontology annotations, functional domains, and evolutionary sequence data to predict subcellular localization of viral proteins in host cells. Using this approach, we predicted the subcellular localization sites of 39 candidate metal-associated MPXV proteins to infer their potential functional roles in host-viral interaction.

Comparative sequence analysis was performed to infer evolutionary relationships among candidate metal-associated MPXV proteins. Protein sequences were aligned using MUSCLE (Multiple Sequence Comparison by Log-Expectation) and analyzed with MEGA11 [57]. A maximum-likelihood phylogenetic tree was constructed under the Jones-Taylor-Thornton (JTT) amino acid substitution model with 500 bootstrap replicates to assess node support. The resulting phylogenetic tree was visualized and annotated using the *ggtree* package in RStudio version 2025.05.0.496 [58].

4.9 | Homology Sequence Search of Identified Putative MBPs With Reference to the Model Organism *Vaccinia*

In the absence of experimentally validated MPXV host-virus protein interaction data, the complete proteome of the Vaccinia virus (VACV) was retrieved from the UniProt reference proteome database (UP000000344) as a comparative model [35]. The VACV proteome consisted of 218 proteins. These sequences were aligned against 39 candidate metal-associated MPXV proteins using standalone BLASTp with an e-value of 1×10^{-5} [59]. Redundant hits were filtered using the CD-HIT at 90 % sequence identity [60]. A total of 45 VACV protein sequences showed homology to the candidate metal-associated MPXV proteins, indicating potential functional conservation of metal-associated roles across orthopoxviruses. These orthologous proteins were further analyzed to infer host-virus interaction mechanism.

4.10 | Network Analysis

The VACV proteins showing greater than 90 percent sequence identity to the candidate metal-associated MPXV proteins were used as query sequences to retrieve human virus-host protein interaction data from VirHostNet 3.0, a curated knowledge base for virus-host protein-protein interaction networks. Interaction data were retrieved in PSI-MITAB 2.5 format via PSICQUIC web services (http://virhostnet.prabi.fr/) [35]. The resulting protein interaction networks were visualized and analyzed using Cytoscape v3.10.4.

4.11 | Structure Validation by Performing PDB Homolog Search and Comparison

To validate the structural reliability of the predicted MPXV metal-associated proteins, all 39 shortlisted proteins were subjected to BLASTp searches against the Protein Data Bank (PDB) database using the NCBI BLAST + platform (https://blast.ncbi.nlm.nih.gov/Blast.cgi) [59]. For each query protein, the most closely related experimentally resolved structure was identified using stringent selection criteria comprising a minimum percentage identity of ≥30% and a query coverage of≥70%. The experimentally derived PDB homologs satisfying the selection criteria were downloaded (.pdb format) for downstream structural alignment and comparison.

Structural superimposition between the experimentally resolved PDB structure and the corresponding AlphaFold3 predicted models was performed using the TM-align server (https://aideepmed.com/TM-align/) [61]. Both structures were uploaded as independent input files, and pair-wise structural alignment was executed using default parameters.

Structural similarity was evaluated primarily using the TM-score metric (0.0 < TM-score < 0.30 means "random structural similarity" and 0.5 < TM-score < 1.0 means "sharing similar structural fold"). Additional alignment parameters gener-

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ated by TM-align server, including aligned length, root mean square deviation (RMSD), and sequence identity, were also considered for structural confidence assessment. RMSD values below 2 Å were interpreted as indicative of excellent structural agreement, whereas higher sequence identity and longer aligned regions further supported structural conservation between predicted and experimentally resolved proteins. Metal-coordinating amino acid residues were also recorded to study the positional and functional conservation in both the AlphaFold3 predicted structure of the candidate MPXV metal-associated protein and its PDB homolog (Supporting Table S3).

The resulting superimposed structural models were downloaded and further visualized using Molstar Viewer, a molecular visualization platform for detailed examination of metal-interacting residues, coordination environments, and overall structural topology. (https://molstar.org/viewer/) [62]. High-resolution three-dimensional images were generated for comparative analysis on UCSF ChimeraX [63]. A two-dimensional ligand interaction diagram of the metal and metal-binding residues within the metalloprotein structure was prepared on LigPlot v.2.3.1 (https://www.ebi.ac.uk/thornton-srv/software/LigPlus/) [64].

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the supplementary material of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Figure S1: Gene Ontology (Biological Processes) Ancestral Chart for DNA Recombination Figure S2: Gene Ontology (Cellular Component) Ancestral Chart for Host Cell Cytoplasm. Figure S3: Gene ontology (Molecular Functions) ancestral chart for metal ion binding. Table S1: AlphaFold3 output table for all the shortlisted metal-binding proteins. The table lists the viral protein IDs and scores (ipTM and pTM) for each metal ion. The protein IDs and score in red color represent the proteins "not-shortlisted" for downstream analysis as per the AlphaFold3 confidence threshold of ipTM > 0.8 and pTM > 0.5. Table S2: (a) The table shows the description of 39-candidate metal-associated MPXV proteins. Including their protein ID, the total number of metal ions that show affinity to bind with the protein molecule, gene name, family, subcellular localization, homologue, and the overall function of the protein. (b)The table shows the description of 39-candidate metal-associated MPXV proteins. Including their protein ID, GO term assignment, protein families, and Cofactors of the protein. Table S3: BLASTp results and TM-align metrics for 39-candidate metal-associated MPXV proteins for Structural Validation. Table S4: KO assignment of 39-candidate metal-associated MPXV proteins.

Supporting File 1: pmic70158-sup-0001-Figure1.pdf.
Supporting File 2: pmic70158-sup-0002-TableS1.docx.
Supporting File 3: pmic70158-sup-0003-TableS2.docx.
Supporting File 4: pmic70158-sup-0004-TableS3.docx.
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